

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
LIDAM Energy Group, CORE, UCLouvain

I am applying for the doctoral position on auction-based support schemes for renewable energy at LIDAM. My field is applied microeconometrics, and I want to bring it to energy market design, a literature argued largely in theory while Europe has run the experiments for a decade. I would work with Professor Bert Willems, whose research with Yueting Yu on discriminatory versus uniform-price auctions connects short-run bidding to long-run investment, and with Quentin Lété on capacity investment under risk. ENERPOL, with Jean Hindriks and François Maniquet, addresses the distributional question my thesis estimates.

My quantitative preparation is the strongest part of my record: 1.0 in Mathematics for Economists, 1.3 in Econometrics, and 1.0 in bachelor mathematics and statistics, where 1.0 is the maximum. My theory training includes advanced microeconomics on strategic thinking and competition with Axel Ockenfels, graded 1.3, whose field is market and auction design. Beyond the curriculum I am working through Jehle and Reny in my own time, and I have calibrated a model in MATLAB.

My master's thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, asked whether a national scholarship programme in Brazil raised formal-sector wages, and for whom. I built the panel from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery. Regional exposure is continuous, so I partitioned it into bins and estimated Callaway and Sant'Anna (2021) group-time effects with doubly-robust inference. Low-exposure regions show a wage gain of 3.04 per cent, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response that two-way fixed effects reports with the opposite sign. I validated it with HonestDiD, a Bacon decomposition and placebos.

I want to ask how support-scheme design determines what is built and at what rent. Since the 2016 state aid framework, member states have adopted technology-neutral, technology-specific and mixed auction designs at different dates, which is staggered adoption of a design rather than of a policy. The AURES II database records the design and outcome of every renewable auction in the EU and UK from 2011 to 2021, so the panel exists. I would estimate how joint bidding shifts the awarded technology portfolio and the rent per megawatt hour against not-yet-treated member states, and test whether design changes were timed to expected cost falls. In ten years I want to be producing the evidence that European support-scheme reform is judged on.

M.Sc. Economics completed September 2026, available from October. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger, pia.pinger@uni-koeln.de, and Jun.-Prof. Dr. Oliver Ruhnau, ruhnau@wiso.uni-koeln.de.

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